How Reinforcement Learning Changes Model Circuits: Testing Refined Local Learning Coefficients

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Abstract

Reinforcement learning with verifiable rewards (RLVR) improves model performance, but the internal changes behind these gains remain difficult to characterize. I propose to combine causal circuit analysis with refined local learning coefficients (LLCs) to investigate whether RLVR strengthens existing computations, recruits them into new pathways, or modifies them. A controlled pilot will support experiments on a small pretrained transformer performing arithmetic. Independent causal tests will evaluate whether geometric measurements identify the timing, location, and type of internal change. The study will assess the value and limitations of LLC as a diagnostic for investigating how reinforcement learning changes models before deployment.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) improves reasoning models using automatically checked answers. Behavioral studies report different limits on these gains under different exploration settings (Yue et al., 2025; Deng et al., 2025). Higher accuracy alone leaves an important question open: does RL strengthen existing computations, reuse them in new ways, or change how the model solves a task? Answering this could help developers investigate capability changes before deployment rather than relying only on benchmark scores.

I will study this question by combining causal tests of model circuits with the local learning coefficient (LLC), a geometric measure of the model's learned solution. My goal is to determine whether refined LLC measurements reliably indicate when and where internal mechanisms change, and whether they help distinguish different kinds of change. A controlled pilot will prepare a focused experiment on a small pretrained language model. The potential societal benefit is better evidence for assessing changes in AI systems before people rely on them.

2 Background

2.1 Related Work

LLC measures how tightly combinations of weights must be constrained to keep loss low, acting like an effective count of constrained directions. Estimators sample nearby weights while favoring low loss (Lau et al., 2025). LLC changes have been linked to transformer development (Hoogland et al., 2025). Refined LLC focuses on components or data; Wang et al. (2025) studied head specialization and circuits with causal interventions. RL studies examined regret geometry (Elliott et al., 2026a) and susceptibilities with activation steering (Elliott et al., 2026b).

An exploratory arithmetic-GRPO project associated LLC with reward changes (Ferrao and Lichkovski, 2025). Mechanistic studies investigated RLVR memorization shortcuts (Yan et al., 2026) and circuit preservation under RL versus supervised fine-tuning (Rojas Nunez et al., 2026). Building on these precedents, I will test whether refined LLC reliably tracks independently verified computational changes during RLVR, including disagreements between the methods.

2.2 Prior Work by the Applicant

My prior research provides experience relevant to this proposal in two areas.

Causal Interventions and Model Steering. I explored refusal mechanisms within the Gemma model family, following Arditi et al. (2024). Using directional ablation and activation addition, I tested how candidate refusal directions influence behavior, including how targeted removal could bypass refusals. This work provided practical experience designing causal interventions and interpreting their effects.

Circuit Discovery and Internal Representations. I reproduced aspects of the GPT-2 Small indirect-object-identification circuit (Wang et al., 2023) using TransformerLens and applied ROME-style causal tracing to factual recall (Meng et al., 2022). I also investigated components that support or suppress outputs in a text-to-image model. This background prepares me for causal analysis across training checkpoints; the proposed work will extend it to LLC estimation.

3 Approach

The study combines controlled training with causal and geometric analyses of saved checkpoints.

3.1 Establishing a Controlled Training Setting

Controlled calibration. Within a fixed pilot budget, I plan to train a small transformer on verifiable algebraic problems. Supervised preparation will establish answer formatting and mixed correct/incorrect samples, followed by RLVR with rewards for correct complete answers. A small set of checkpoints and candidate carry components will test intervention reproducibility, LLC stability, and measurement cost before the pretrained-model experiment.

Pretrained-model experiment. I will select a small pretrained, open-weight transformer based on arithmetic ability, intervention and LLC support, and cost. A pilot will identify a short arithmetic task with a unique answer format and room for improvement, such as addition in an unfamiliar numeral base. Related tasks and carry/no-carry problems will probe shared computations. Separate calibration problems will check for enough mixed-reward groups for GRPO (Shao et al., 2024), with brief supervised preparation if needed. I will analyze the original model, prepared checkpoint, and RLVR checkpoints, keeping the training distribution fixed.

Controls and resources. RLVR and supervised fine-tuning will branch from the same prepared checkpoint. Comparisons at similar performance and computational budgets will be reported separately. Pilot runtimes will set checkpoint and component budgets, targeting three independent runs per condition. Regular measurements will be supplemented near candidate changes. H2 is the primary deliverable; classifying change types is exploratory. The controlled study provides a fallback if pretrained-model analysis exceeds available resources.

3.2 Combining Causal and Geometric Analysis

Causal characterization. At regular checkpoints, activation patching, ablations, and path patching will test how attention heads and feed-forward components contribute to digit and carry operations. Attribution patching will screen candidates for exact tests (Syed et al., 2023). Separate discovery/evaluation problems, alternative replacements, and random component controls will assess reliability (Zhang and Nanda, 2024).

Geometric measurements. Following Elliott et al. (2026a), for fixed inputs and one accepted complete answer $y^*(x)$, define expected regret relative to reward one as

$$R(\theta) = E_x[1 - p_\theta(y^*(x) \mid x)].$$

Here p_θ includes termination, so R measures expected failure under the model's sampling distribution. LLC estimates using this objective will be compared with negative log-likelihood; both differ from the GRPO training objective. Component estimates vary selected weights while freezing others, using training and related tasks to investigate specialization. Multiple chains, sampler settings, and loss traces will test whether checkpoint differences exceed estimation noise; unstable estimates will be inconclusive.

4 Evaluation

H2 is the primary test; H1 states an expected mechanistic pattern and H3 is exploratory.

H1: RLVR mainly strengthens or recruits existing arithmetic mechanisms. Strengthening means greater influence on the same operation; recruitment means a component with an established related-task role joins the target operation or a different pathway. Input contrasts and component/path interventions will distinguish these cases, reporting unresolved cases separately. Larger ablation effects alone are insufficient; an undetected baseline role does not establish absence.

H2: LLC changes identify when and where causal roles change. The pilot will fix detection rules and a checkpoint matching window. Independently detected LLC and intervention changes will be compared across time, components, and runs, reporting matches, missed causal changes, and unmatched LLC signals. Reward-based signals and evenly spaced inspections provide baselines; shifted timelines test alignment beyond shared training trends.

H3 (exploratory): Task-specific LLC patterns differ between recruitment and strengthening. Recruitment and strengthening may produce distinct LLC patterns across training and related tasks. For causally established cases, I will test these differences on separate runs, supported by accuracy, answer probabilities, and carry/no-carry performance. H2 remains assessable without this distinction.

5 Discussion

I expect partial agreement between geometric LLC measurements and causal interventions, with temporal alignment proving easier to identify than structural change types. If LLC shifts reliably track circuit changes, geometric tracking can serve as an efficient screening tool to guide targeted path patching; if agreement is weak, it highlights the boundaries of LLC as an RLVR diagnostic. While these findings initially apply to short arithmetic tasks, releasing all code, checkpoints, and failure cases will help establish standardized pre-deployment audit practices for broader reasoning models.

6 Conclusion

This project asks how RLVR changes existing computations and whether refined LLC can help identify those changes. By combining controlled training, causal tests, and geometric measurements, it will evaluate a potential tool for understanding what reinforcement learning changes before a model is deployed.

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